Stage 1 Hard Drill — With Solutions

Chapter 6.4 — Optimal Capital Growth & Position Sizing

10 new competition-grade questions, full-chapter coverage, full worked solutions. Target: under 90 seconds each once the shortcut is spotted.

Q1 — A Bet That Can Push

A bet pays +1 per unit staked with probability 0.5, loses the stake with probability 0.4, and with probability 0.1 is a *push* (stake returned). Find the Kelly fraction and the growth rate. What goes wrong if you treat the push as a loss?

Solution. With fraction f staked, wealth is multiplied by $1+f$ (prob. 0.5), $1-f$ (prob. 0.4), or 1 (prob. 0.1, contributes $\ln 1 = 0$):

$$G(f) = 0.5\ln(1+f) + 0.4\ln(1-f)$$

$$G'(f)=0: \quad 0.5(1-f) = 0.4(1+f) \;\Rightarrow\; f^\star = \frac{p-q}{p+q} = \frac{0.1}{0.9} = \boxed{\tfrac{1}{9}}$$

$$G^\star = 0.5\ln\tfrac{10}{9} + 0.4\ln\tfrac{8}{9} = 0.05268 - 0.04711 = \boxed{0.00557}$$

This equals the Kelly fraction for the *conditional* win probability $p' = \frac{0.5}{0.9} = \frac{5}{9}$ ($f = 2p'-1 = \frac{1}{9}$), with growth scaled by 0.9, the fraction of bets that are decided.

The trap: treating the push as a loss gives $q = 0.5$ and $p = 0.5$, so $f^\star = 0$ ("no edge") — a wrong answer. A push changes the *speed* of growth, not the optimal size.

⋆ *Intuition:* a push is a bet that never happened. Size by the *decided* outcomes only.

Q2 — Betting Into a Commission

An even-money bet wins with probability 0.55, but the house keeps 5% of every *winning* payout (a win pays 0.95 per unit). Find the Kelly fraction, the growth rate versus the commission-free case, and the break-even win probability.

Solution. Net odds $b = 0.95$.

$$f^\star = \frac{bp-q}{b} = \frac{0.95(0.55)-0.45}{0.95} = \frac{0.0725}{0.95} = \boxed{0.0763}$$

$$G^\star = 0.55\ln(1+0.95f) + 0.45\ln(1-f) = \boxed{0.00277}$$

Without commission: $f^\star = 0.1$ and $G^\star = 0.00501$. The 5% vig removes 45% of the growth. If you ignore the vig and still bet 0.1:

$$G(0.1) = 0.55\ln 1.095 + 0.45\ln 0.9 = 0.00250$$

which is 90% of the optimum. Break-even: $bp = q \iff p = \frac{1}{1+b} = \frac{1}{1.95} = \boxed{51.28\%}$.

▶ *Speed trick:* costs reduce the *edge* $bp-q$ one-for-one but also shrink the denominator; always recompute b net of fees before sizing.

Q3 — A Three-Horse Race

Three horses win with probabilities $(0.5, 0.3, 0.2)$. A winning bet of 1 on a horse returns o in total (stake included). (*a*) Odds $o = (3,4,6)$: find the Kelly split and growth. (*b*) Odds $o = (2.4, 2.9, 4.0)$: find the Kelly portfolio.

Solution. (*a*) $\sum \frac{1}{o} = \frac{1}{3}+\frac{1}{4}+\frac{1}{6} = 0.75 < 1$: the book is generous, so bet everything. Maximise $\sum p_i \ln(f_i o_i)$ subject to $\sum f_i = 1$; the Lagrange condition $\frac{p_i}{f_i} = \lambda$ gives

$$f_i = p_i = (0.5, 0.3, 0.2)$$

Wealth multipliers are $p_i o_i = (1.5, 1.2, 1.2)$:

$$G = 0.5\ln 1.5 + 0.3\ln 1.2 + 0.2\ln 1.2 = \boxed{0.2939}$$

(*b*) $\sum \frac{1}{o} = 1.011 > 1$: the book takes a margin, so keep a cash reserve R. With $p_i o_i = (1.2, 0.87, 0.8)$, include horses in order of $p_i o_i$ while $p_i o_i > R$, where

$$R = \frac{1-\sum_S p_i}{1-\sum_S 1/o_i}$$

$S=\{1\}$: $R = \frac{0.5}{0.5833} = 0.857 < 0.87$, so horse 2 also qualifies. $S=\{1,2\}$: $R = \frac{0.2}{0.2385} = 0.8386 > 0.8$, so horse 3 is excluded. Stakes $f_i = p_i - \dfrac{R}{o_i}$:

$$f_1 = 0.5 - \tfrac{0.8386}{2.4} = \boxed{0.1506}, \quad f_2 = 0.3 - \tfrac{0.8386}{2.9} = \boxed{0.0108}$$

Cash $= R = 0.8386$ (stakes sum to 0.1614). Growth 0.01417, slightly above the single-horse Kelly bet ($f = 0.1429$ on horse 1 only, $G = 0.01409$). A numerical optimiser confirms these stakes.

⋆ *Intuition:* R is the *price of cash*. Horse 2 has a negative edge ($0.87 < 1$) but still earns a stake because it hedges the case where horse 1 loses.

Q4 — Sizing Under a Drawdown Budget

Even-money bets with $p = 0.6$ (full Kelly $f^\star = 0.2$). You want $\mathbb{P}$(ever falling to 50% of capital) $\le 10\%$. What fraction of Kelly should you bet and what share of the maximum growth is kept?

Solution. Use the drawdown law $\mathbb{P} \approx \alpha^{2/c-1}$ with $\alpha = \frac{1}{2}$:

$$\left(\tfrac{1}{2}\right)^{2/c-1} \le 0.1 \;\Rightarrow\; \frac{2}{c} - 1 \ge \log_2 10 = 3.322$$

$$c \le \frac{2}{4.322} = \boxed{0.463} \;\Rightarrow\; f = 0.463(0.2) = 0.0926$$

Growth kept $= c(2-c) = 0.463(1.537) = \boxed{71\%}$.

Check with the actual discrete bets: solving $p(1+f)^{-\theta} + q(1-f)^{-\theta} = 1$ for $\theta = 3.322$ gives $f = 0.0934$ ($c = 0.467$, 71.3% of growth). A simulation of 6×10^4 paths gives $\mathbb{P}(\text{ever} \le 50\%) = 8.6\%$ at $f = 0.0926$ and 8.8% at $f = 0.0934$: the diffusion rule is slightly conservative.

▶ *Speed trick:* invert $\alpha^{2/c-1} = \varepsilon$ as $c = \dfrac{2}{1+\log_{1/\alpha}(1/\varepsilon)}$. A 10% risk of a 50% drawdown costs about 29% of the growth.

Q5 — Leverage With a Borrowing Spread

An asset has excess return $\mu = 6\%$ over cash and volatility $\sigma = 15\%$. Borrowing (any leverage above 1) costs an extra 2% per year over cash. Find the optimal leverage and growth, and compare with the unconstrained Kelly answer.

Solution. Unconstrained: $f^\star = \frac{\mu}{\sigma^2} = \frac{0.06}{0.0225} = 2.67$ with $g^\star = \frac{\mu^2}{2\sigma^2} = 0.08$.

With the spread $s = 0.02$ on the borrowed part $(f-1)$:

$$g(f) = f\mu - (f-1)s - \tfrac{1}{2}f^2\sigma^2 \quad (f > 1)$$

$$g'(f) = \mu - s - f\sigma^2 = 0 \;\Rightarrow\; f^\star = \frac{\mu - s}{\sigma^2} = \frac{0.04}{0.0225} = \boxed{1.78}$$

$$g(1.78) = 0.1067 - 0.0156 - 0.0356 = \boxed{5.56\%}$$

Check other leverages: $g(1) = 4.88\%$, $g(2) = 5.50\%$, $g(2.67) = 4.67\%$, $g(3) = 3.88\%$. Using the unconstrained 2.67 would waste about 0.9 points per year.

⋆ *Intuition:* borrowing shifts the effective expected excess return above leverage 1 from μ to $\mu - s$. The Kelly leverage is computed with the *marginal* cost of the last dollar.

Q6 — Two Correlated Assets

Two assets have excess returns $\mu = (6\%, 4\%)$, volatilities $(20\%, 15\%)$ and correlation $\rho = 0.5$. Find the Kelly portfolio and growth. Compare with the uncorrelated case and with putting each asset's separate Kelly size together.

Solution. Covariance matrix: $\Sigma = \begin{pmatrix} 0.04 & 0.015 \\ 0.015 & 0.0225 \end{pmatrix}$ (the off-diagonal is $0.5 \cdot 0.2 \cdot 0.15$). The Kelly vector maximises $f^\top\mu - \frac{1}{2}f^\top\Sigma f$:

$$f^\star = \Sigma^{-1}\mu, \qquad g^\star = \tfrac{1}{2}\mu^\top\Sigma^{-1}\mu$$

$\det\Sigma = 0.0009 - 0.000225 = 0.000675$:

$$f_1 = \frac{0.0225(0.06) - 0.015(0.04)}{0.000675} = \boxed{1.111}$$

$$f_2 = \frac{-0.015(0.06) + 0.04(0.04)}{0.000675} = \boxed{1.037}$$

$$g^\star = \tfrac{1}{2}(1.111 \cdot 0.06 + 1.037 \cdot 0.04) = \boxed{5.41\%}$$

Single-asset Kelly growths are 4.5% and 3.56%. If the assets were uncorrelated, $f = (1.5, 1.78)$ and $g = 8.06\%$. Naively using those uncorrelated sizes on the correlated assets gives only 4.06% (a leveraged portfolio with volatility about 49%).

▶ *Speed trick:* in the multivariate Kelly problem correlation *reduces* the benefit of diversification and the sizes shrink. Always solve $\Sigma f = \mu$.

Q7 — Ruin With Unequal Steps

A bankroll changes by $+2$ or -1 with probability $\frac{1}{2}$ each, per round, starting from k units; ruin means reaching 0. Find $\mathbb{P}(\text{ever ruined})$ from k and the smallest k giving ruin $\le 1\%$.

Solution. The drift is $+\frac{1}{2} > 0$, so ruin is not certain. Let $\psi(k) = \theta^k$:

$$\psi(k) = \tfrac{1}{2}\psi(k+2) + \tfrac{1}{2}\psi(k-1) \;\Rightarrow\; \theta = \tfrac{1}{2}\theta^3 + \tfrac{1}{2}$$

$$\theta^3 - 2\theta + 1 = 0 = (\theta - 1)(\theta^2 + \theta - 1)$$

The root $\theta = 1$ is trivial; the admissible root is

$$\theta = \frac{\sqrt{5} - 1}{2} = 0.618$$

(the golden ratio conjugate). Since down-steps are exactly 1, the walk cannot jump past 0, so

$$\boxed{\mathbb{P}(\text{ruin} \mid k) = \left(\tfrac{\sqrt{5}-1}{2}\right)^k}$$

From $k = 5$: 0.0902 (simulation 0.0909). Need $0.618^k \le 0.01$: $k \ge \frac{\ln 0.01}{\ln 0.618} = 9.57$, so $k = 10$ (ruin 0.81%).

⋆ *Intuition:* with equal-size steps the ruin probability is $(q/p)^k$; with unequal steps the ratio q/p is replaced by the root θ of the one-step moment equation $\mathbb{E}[\theta^{-X}] = 1$.

Q8 — Bold Play or Timid Play?

Roulette-style even-money bets win with probability 0.47. You hold 50 and want to reach 100. Compare staking the whole 50 at once (bold play) with staking 1 at a time, 10 at a time, and 25 at a time.

Solution. Bold play: one bet decides everything, so $\mathbb{P} = \boxed{0.47}$.

Staking m per bet turns the problem into units: start $k = \frac{50}{m}$, target $N = \frac{100}{m}$, with $r = \frac{q}{p} = \frac{0.53}{0.47} = 1.1277$ and

$$h = \frac{1 - r^k}{1 - r^N} = \frac{1}{1 + r^k} \quad (N = 2k)$$

$$m = 1: \; k = 50, \; r^{50} = 406 \;\Rightarrow\; \mathbb{P} = 0.0025$$

$$m = 10: \; k = 5, \; r^5 = 1.823 \;\Rightarrow\; \mathbb{P} = 0.354$$

$$m = 25: \; k = 2, \; r^2 = 1.272 \;\Rightarrow\; \mathbb{P} = 0.440$$

So $0.0025 < 0.354 < 0.440 < 0.47$: the bigger the stake, the better. A dynamic-programming check on a smaller board (10 target from 5) confirms 0.47 as optimal.

▶ *Speed trick:* in an *unfavourable* game, to maximise the chance of reaching a target, bet as boldly as possible (Dubins–Savage). In a *favourable* game, the opposite (small, Kelly-style bets) is right.

⋆ *Intuition:* when the house has the edge, time is your enemy: each extra bet gives the edge another chance to grind you down.

Q9 — Is the Coin Biased? Kelly Under Parameter Uncertainty

An even-money bet has unknown win probability: $p = 0.5$ or $p = 0.7$, each with prior $\frac{1}{2}$. (*a*) What is the Kelly fraction? (*b*) How does it change after the observed outcomes W, WW, WL? (*c*) What is the average growth if you bet as if $p = 0.7$, i.e. $f = 0.4$?

Solution. (*a*) G is linear in p for fixed f, so maximising the average of G over the prior is the same as using the *posterior mean* $\bar{p}$:

$$\bar{p} = \tfrac{1}{2}(0.5) + \tfrac{1}{2}(0.7) = 0.6, \qquad f^\star = 2\bar{p} - 1 = \boxed{0.2}$$

Check: $G(0.2) = \frac{1}{2}(-0.0204) + \frac{1}{2}(0.0607) = 0.0201 = G(0.2; p{=}0.6)$.

(*b*) Bayes: $\mathbb{P}(0.7 \mid \text{data}) \propto 0.7^w 0.3^l$ versus 0.5^{w+l}:

$$\text{W}: \; \tfrac{0.35}{0.35+0.25} = 0.583, \; \bar{p} = 0.617, \; f = \boxed{0.233}$$

$$\text{WW}: \; \tfrac{0.49}{0.49+0.25} = 0.662, \; \bar{p} = 0.632, \; f = \boxed{0.265}$$

$$\text{WL}: \; \tfrac{0.21}{0.21+0.25} = 0.457, \; \bar{p} = 0.591, \; f = \boxed{0.183}$$

(*c*) At $f = 0.4$: $G(p{=}0.5) = -0.0872$ and $G(p{=}0.7) = +0.0823$:

$$\mathbb{E}[G] = \tfrac{1}{2}(-0.0872) + \tfrac{1}{2}(0.0823) = \boxed{-0.0024 < 0}$$

Betting as if the optimistic hypothesis were true has *negative* expected growth.

⋆ *Intuition:* the Kelly bettor trusts the *average* belief, not the best-case belief, and updates the stake as evidence arrives.

Q10 — Capstone: How Much of an Estimated Sharpe Ratio Can You Trust?

A strategy has estimated Sharpe ratio $\hat{S} = 1.0$ from $T = 4$ years of data (standard error $\approx 1/\sqrt{T} = 0.5$, true Sharpe $\sim \mathcal{N}(1, 0.5^2)$). You lever it to c times the Kelly leverage computed from $\hat{S}$. (*a*) Express the true growth rate in terms of $r = S_{\text{true}}/\hat{S}$. (*b*) Find $\mathbb{P}(\text{growth} < 0)$ and the expected growth for $c = 1, \frac{1}{2}, \frac{1}{4}$. (*c*) What is the Kelly leverage at volatility 10%?

Solution. (*a*) With true excess return $\mu_t = r\hat{\mu}$ and leverage $f = c\hat{\mu}/\sigma^2$:

$$g = f\mu_t - \tfrac{1}{2}f^2\sigma^2 = \frac{\hat{\mu}^2}{\sigma^2}\left(cr - \frac{c^2}{2}\right) = \hat{S}^2\, c\left(r - \frac{c}{2}\right)$$

Growth is negative iff $r < \frac{c}{2}$.

(*b*) $r \sim \mathcal{N}(1, 0.5^2)$:

$$c = 1: \; \mathbb{P}(r < 0.5) = \Phi(-1) = \boxed{15.9\%}$$

$$c = \tfrac{1}{2}: \; \mathbb{P}(r < 0.25) = \Phi(-1.5) = \boxed{6.7\%}$$

$$c = \tfrac{1}{4}: \; \mathbb{P}(r < 0.125) = \Phi(-1.75) = \boxed{4.0\%}$$

Expected growth: $\mathbb{E}[g] = \hat{S}^2 c(1 - \frac{c}{2}) = \frac{\hat{S}^2}{2}\, c(2 - c)$, i.e. 100%, 75%, 43.8% of $\frac{\hat{S}^2}{2} = 0.5$ per year.

(*c*) $\hat{\mu} = \hat{S}\sigma = 0.10$: $f^\star = \dfrac{\hat{\mu}}{\sigma^2} = \dfrac{0.10}{0.01} = \boxed{10\times}$ leverage; half-Kelly is $5\times$.

Decision: half-Kelly keeps 75% of the expected growth and cuts the chance of losing money from 15.9% to 6.7%. Going below $\frac{1}{4}$ Kelly barely reduces the risk further (the floor is the chance that the true Sharpe is ≤ 0) while giving up more than half of the growth.

⋆ *Full-circle intuition:* an estimated Sharpe ratio is a noisy speedometer. Kelly tells you the speed limit for the *true* road; fractional Kelly is the margin you take because you can only see the road through noisy glasses.